

Junghun Hwang (황정헌)

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Education

Konkuk University

Mar 2022 - Expected Feb 2027

B.S. in Smart Mobility Engineering | Seoul, Korea

Completed coursework: Linear Algebra, Autonomous System Platform, Artificial Intelligence Applications, Data Structures and Algorithms

In progress: AI Mobility Systems

Research Interests

Autonomous-driving perception and decision-making, with an emphasis on learning-based and end-to-end policies and their closed-loop behavior under occlusion and input delay.

Research & Project Experience

Kookmin Autonomous Driving Competition - AI-Edge Challenge

Apr 2026 - Aug 2026

Team Lead | Team of 5

- Built and integrated a ROS 2 steering pipeline spanning data collection, imitation-learning training, TorchScript inference, and switching between learned and rule-based control.
- Prepared and reviewed Roboflow datasets and trained a unified YOLO detector for traffic lights, vehicles, obstacles, and cones.
- Integrated mission-management and driving modules and tuned the competition vehicle. The team used rule-based control in the final competition after persistent oscillation in the learned steering policy.
- Defined parking waypoints on a SLAM map and implemented reverse recovery logic for the parking event.

Team results: **Preliminary 2nd Place** | Finals 13th Place (course completed) | **Parking Special Award**

2026 Student AI · SW Mobility Competition

Mar 2026 - Ongoing

Team of 27 | Decision-Making Subteam (5)

- Demonstrated real-vehicle waypoint driving at **10 m/s (36 km/h)** on both preliminary and final courses during venue practice on Sep 19-20, 2026.
- Developed ROS 2 cone-boundary path generation, red-cone-triggered stop commands, and fail-safe logic; implemented Pure Pursuit–Stanley tracking with steering constraints.
- Compared hybrid tracking with Stanley alone in matched Gazebo segments at 6 m/s; reduced 95th-percentile heading error by approximately **14%** and steering-command total variation by approximately **10%**.
- Authored the high-level control section of Team K.A.I.'s competition technical report, documenting hybrid path tracking, steering constraints, and vehicle-command integration.
- Confirmed red-cone-triggered AEB stop commands in a 50 km/h vehicle trial; a brake hardware fault prevented an in-zone stop.

Camera-Based V2I Cooperative End-to-End Autonomous Driving

2026 - Ongoing

Capstone Design | Requirements & System Design; **Implementation Planned**

- Designed a matched comparison of camera-only and V2I-assisted end-to-end driving under pedestrian occlusion, using a 1/5-scale HENES platform in daytime, low-speed conditions.
- Specified a road-plane homography pipeline and direct transmission of pedestrian position, confidence, and timestamps for lightweight fusion into waypoint and target-speed outputs.
- Formulated a detection-distance budget from latency, braking distance, and a safety margin. Planned measurements of stagewise latency, earlier pedestrian detection and deceleration, minimum TTC, and collisions.

UAV Waypoint Mission - ROS 2 / PX4 SITL

May 2026 - Jun 2026

Autonomous System Platform Course | UAV Waypoint Development | Team of 4

- Developed a ROS 2 mission package with CSV waypoints, TF-based arrival checks, and coordinate-mode handling; integrated it with an existing PX4 offboard controller via PoseStamped commands.
- Demonstrated sequential waypoint tracking and final autonomous landing in the PX4 SITL/Gazebo course simulation.

ROS 2 / CARLA Decision and Control

Mar 2026 - Jun 2026

Dream Semester Project | Decision-Making | Team of 3

- Implemented ROS 2 decision-making, Pure Pursuit waypoint tracking, and longitudinal control with CARLA integration and later vehicle-interface adaptation.
- Investigated simulated tracking instability by adjusting steering direction, lookahead, smoothing, and command limits; instability remained unresolved, with no real-vehicle run.

Awards & Competitions**2026 Kookmin AI-Edge Challenge**

Parking Special Award (team)

2022 Drone Clash, General Division

3rd Place

Technical Skills**Languages:** Python, C++**Autonomous Systems:** ROS 2, CARLA, Gazebo, PX4 SITL**AI / Computer Vision:** PyTorch, YOLO, OpenCV, Roboflow**Tools / Environment:** Ubuntu, Git/GitHub, Foxglove, RViz2, rosbag2, QGroundControl